Lecture1 (Unit 1.1). Theme 1

Wednesday, 11 January 2023 16:21

Source: Stewart, Appendix A pages A2 - A6

Numbers, Intervals & Inequalities

Note

1. Writing mathematics correctly is very important
2. Notation and symbols must be used correctly.

Number systems

$\mathbb{N}$ – natural numbers

$\mathbb{Z}$ – Integers

$\mathbb{Q}$ – Rational numbers

$\mathbb{I}$ – Irrational numbers

$\mathbb{R}$ – Real numbers

Symbols and notation	Meaning
• $\Rightarrow$ • :	implies
• $\sim$ • :	equivalent
• $=$:	equal to
• $\cup$:	union (or)
• $\cap$:	Intersection (and)

Intervals

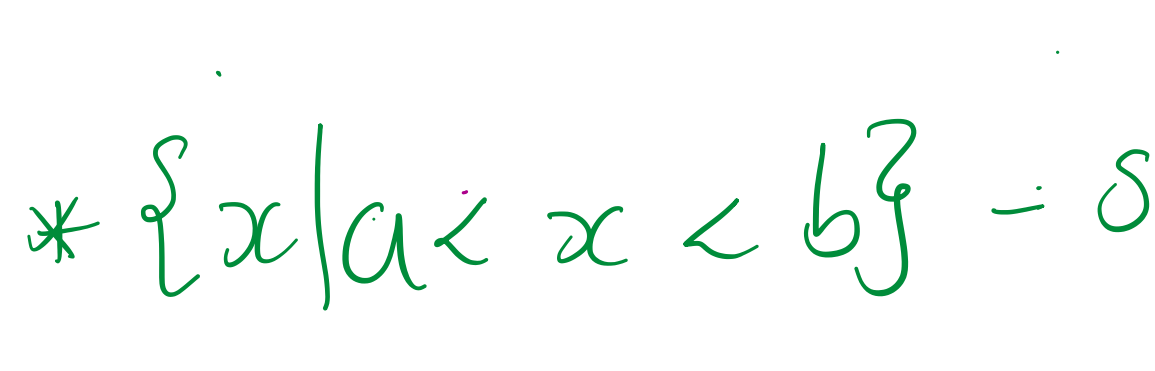
Set of real numbers

• Open interval

* (a, b) – interval notation *

* $\{x | a < x < b\}$ – set notation

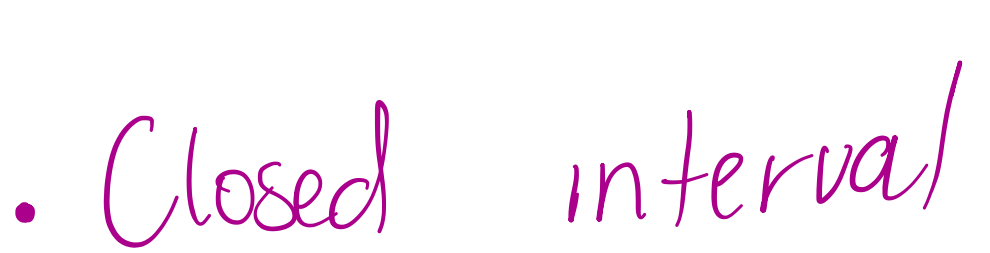
Geometric representation



• Closed interval

– $[a, b]$

– $x|$ - - - - }



See: Table of Intervals : A4

Inequalities

Symbols: $>$, $<$, $\le$, $\ge$

* Solution of an inequality is a set of values that make the inequality true.

See: Rules of Inequalities (A4)

Solving Inequalities

Solve the following inequalities and write your answers in interval notation

Examples:

1. $3x - 11 < 4$

2. $2x + 1 < 5x - 8$

3. $-5 \le 3 - 2x \le 9$ (22)

Note: Always use a graph to solve inequalities.

Solve for x:

4. $\frac{x+1}{2-x} \ge 0$

5. $x^2 + x < 2$

6. $-x^2 - x - 1 < 0$

END.

7. $\frac{1}{x+1} > 1$

8. $x^2 > 4$

9. $x^2 < 4$